Career PathFinder

Grounded, Adaptive & Explainable AI Learning Path Recommender Engine

Competition: HCLTech AMPlified S1, Round 2	**Live Demo URL:** https://career-pathfinder-nika619.onrender.com/
Team LODESTAR: Mayank Tiwari (Leader), Shubh Yadav, Uma Jaiswal	**GitHub Repo:** https://github.com/nika619/Career-pathfinder-nika619
Institution: Babu Banarasi Das University (B.Tech 2023–2027)	**Core Stack:** FastAPI, NetworkX, SQLite WAL, Three.js 3D WebGL

1. Executive Summary & Problem Statement

Modern upskilling platforms often deliver ungrounded, hallucinated learning lists that ignore prerequisite hierarchies and fail to measure true prior knowledge. **Career PathFinder** is a deterministic, mathematically grounded engine that inverts this paradigm: *the mathematical engine computes the exact prerequisite roadmap and Bayesian knowledge vectors, while LLM models serve purely to phrase explanations.*

Key Differentiators:

• **Real 53-Node Prerequisite DAG:** 61 strictly validated dependency edges solved monotonically via NetworkX Kahn's Topological Sort.
• **Bayesian Knowledge Tracing (BKT):** Adaptive cognitive assessment updating prior mastery probabilities dynamically without blind self-reports.
• **O*NET 2024.2 Labor-Market Grounding:** Direct gap quantification against 9 standardized industry career roles.
• **Glass-Box Explainability:** Complete mathematical, labor-market, and prerequisite justification for every sequenced milestone.
• **Multi-Agent AI Copilot:** 4 distinct domain mentor personas (General Coach, Algo Specialist, System Architect, Career Strategist) with voice synthesis, Mermaid architecture diagrams, and 60s micro-challenges.

2. Mathematical & Algorithmic Architecture

Engine Module	Mathematical / CS Foundation	Operational Impact
Prerequisite DAG Engine	Directed Acyclic Graph G=(V, E) Topological Sort (Kahn's Algorithm)	Guarantees zero prerequisite violations; entry-point foundational skills strictly precede advanced concepts.
Cognitive Diagnostic (BKT)	Bayesian Knowledge Tracing: P(L_t|Obs) = P(L_{t-1})·P(Obs|L) / P(Obs) P(L_t) = P(L_t|Obs) + (1-P(L_t|Obs))·P(T)	Dynamically measures true prior competencies using slip (0.10) and guess (0.20) parameters, avoiding inaccurate self-reporting.
Labor Market Gap Analyzer	O*NET Vector Distance: Gap(s) = I(s, Role) · (1 - M(s))	Calculates true competency deficiency weighted by government labor-market importance indices (0–100%).
Adaptive Re-Planner	Dynamic Subgraph Re-computation on Pass / Fail Trigger	Splits milestones into remedial bridge modules on failure or fast-tracks mastered competencies instantly.

3. End-to-End System Architecture

The application is structured into a modular, production-grade micro-architecture backed by FastAPI and SQLite WAL persistence:

Layer 1: Spatial UI & 3D Interactive Client
• Vanilla ES6+ WebGL with Three.js 3D background constellation, synaptic filaments, and orbital polyhedral core.
• Dual-Deck Command Switcher: Seamlessly toggle between AI Copilot and Glass-Box Trace Inspector without vertical scroll fatigue.
• Mermaid.js dynamic visual architecture diagram generator and Web Speech API voice synthesis.

Layer 2: REST API & Routing (FastAPI)
• POST /api/goal — Natural-language career ambition parsing & role resolution.
• POST /api/diagnostic/next & /api/diagnostic/answer — Adaptive BKT question loop & posterior updates.
• POST /api/plan — Deterministic NetworkX topological sorting & roadmap generation.
• POST /api/explain — Mathematical & labor-market glass-box trace inspection.
• POST /api/chat — Multi-agent AI mentor copilot grounded strictly in user state.
• GET /api/graph — Complete 53-node DAG topology for interactive visualization.

Layer 3: Core Algorithmic Engine & Persistence
• dag_engine.py: Pre-compiled NetworkX DAG with cycle detection and monotonic topological sorting.
• diagnostic.py: 4-parameter Bayesian Knowledge Tracing engine with fuzzy answer normalization.
• database.py: Thread-safe SQLite WAL persistence with automatic schema migration and transaction safety.

4. UI/UX Innovations & User Experience

The interface was crafted to deliver an extraordinary, state-of-the-art interactive experience:

• **Dual-Deck Command Cockpit:** Solves vertical scroll fatigue by hosting the AI Copilot and Glass-Box Trace in a synchronized tabbed deck with viewport-capped internal scrolling.
• **Pinned Chat Control Dock:** AI chat input and voice controls are pinned with zero clipping across all screen resolutions.
• **Interactive Prerequisite Clarity:** Clean contextual badges clearly distinguishing between foundational entry points and met prerequisite dependencies.
• **3D Readiness Hologram Gauge:** Orbiting 3D rings with real-time Bayesian mastery vectors calculated per milestone.
• **Outcome-Based Capstone Evaluator:** 4-point structured rubric modal with code repository inspection and automated scoring (0–100 Pts).
• **Multi-Format Syllabus Export:** One-click syllabus export to clean PDF, Markdown (Notion/Obsidian), or structured JSON.

5. Quality Assurance & Test Verification

Test Domain	Test Cases & Coverage	Result
DAG Topological Sort	Acyclic verification, 61 dependency edge constraints, monotonic difficulty progression (1→4).	**PASSED (100%)**
Bayesian BKT Engine	Prior probability updates, posterior surge on correct answers (0.200 → 0.609), slip/guess bounds.	**PASSED (100%)**
Fuzzy Answer Normalizer	Handles trailing whitespace, quotes, uppercase/lowercase variations in diagnostic verification.	**PASSED (100%)**
Deterministic Fallbacks	100% operation of all algorithms when external LLM API keys are absent or rate-limited.	**PASSED (100%)**
Live Cloud Deployment	Render Docker deployment, SQLite WAL persistence, sub-50ms API endpoint response times.	**PASSED (100%)**

6. Project Sign-Off & Team Credentials

■ **Team Name:** LODESTAR | **Institution:** Babu Banarasi Das University (B.Tech 2023–2027)
■ **Team Members: Mayank Tiwari** (Leader), **Shubh Yadav** (Member), **Uma Jaiswal** (Member)
■ **Event:** HCLTech AMPlified S1, Round 2 — AI-Powered Personalized Learning Path Recommender Engine